Lab brief: does mass slow the change?

Same driving force. Different masses. Your data decides.

11 A

TASK 1 · CONTROLS CHECK

- ☐ hanging mass stays the same
- ☐ track is level
- ☐ start from rest, same spot
- ☐ release with no push
- ☐ counted cart + added + hanger
- ☐ 3 trials per mass

TASK 2 · MY DATA

TOTAL MASS (kg)	TRIAL 1 a	TRIAL 2 a	TRIAL 3 a	AVERAGE a	m × a

TASK 3 · GRAPH: AVERAGE a vs. MASS

TASK 4 · CHECK THE PRODUCT

Is m × a about the same in every row? What does that number represent?

ANALYST SIGN-OFF · CLAIM FROM MY DATA

When system mass increased, acceleration ____. Evidence: ____ vs. ____.

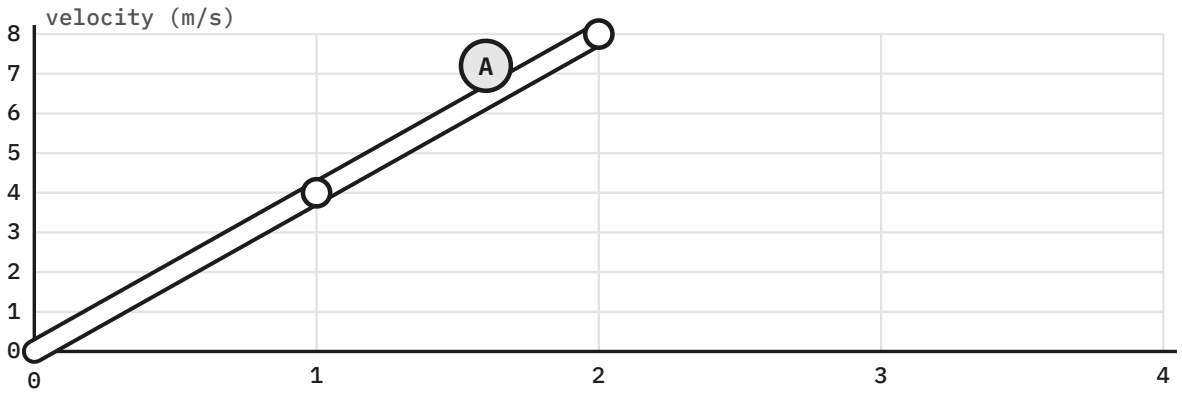
S3 1 2 3

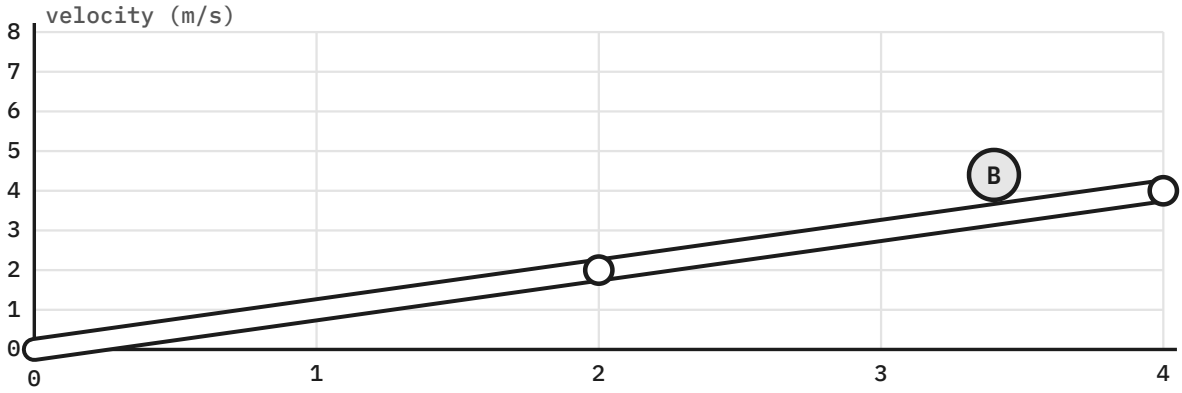
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Read the sample lab graph

11 F

DIRECTIONS · Illustrative data (not yours): same 4 N net force, different masses.





A · 1 kg CART

slope a = ____ m/s²

B · 4 kg CART

slope a = ____ m/s²

PATTERN

mass × 4 → a × ____

EXPLAIN THE PATTERN

Use inertia: why does the heavier cart gain velocity more slowly?

S3 1 2 3

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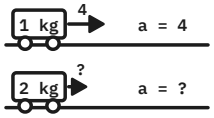
UNIT 1 · DAY 11 · S3 · SEI · PARTNER TALK

Talk it through: the mass lab

Language goal: I can **make a claim** from data.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · 1 kg → 4 m/s², 2 kg → ? (SAME FORCE)



A ASKS "What happens to a when the mass doubles?"

B SAYS "The mass goes from ____ to ____ kg, so a goes from 4 to ____ m/s²."

A CHECKS "So you're saying more mass gives ____ acceleration."

ROUND 2 · SWITCH · WHAT STAYED THE SAME?



stays the same

B ASKS "Why did we keep the hanging mass the same?"

A SAYS "To keep the ____ force the same. Then only the ____ changes."

B ADDS "I want to add that the hanger is part of the ____ mass."

ROUND 3 · BOTH · CHECK m × a

1	4
2	2
4	1

×2 ÷2

A "1 × 4 = ____ and 2 × 2 = ____."

B "The product stays ____, so it equals the ____ force."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

When mass increases, acceleration…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

S3 (1) (2) (3)

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UNIT 1 · DAY 11 · S3 · SEI · SENTENCE FRAMES

Say it, then write it: lab claim

Language goal: I can **make a claim** from data.



mass
masa · massa
amount of matter (kg)



control
control · controle
what stays the same



trial
prueba · ensaio
one run of the test



pattern
patrón · padrão
what repeats in data

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: mass · control · trial · decreased · evidence

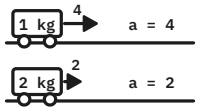
In each ____, we changed only the ____. The hanging mass was the ____. When mass went up, acceleration ____. My ____ is two measured values.

2 · SAY IT TO A PARTNER, THEN WRITE IT



When the mass was ____ kg, the acceleration was ____ m/s².
When mass increased, acceleration ____ because ____ .

3 · SOLVE WITH THE STEPS



Sample: 1 kg → 4 m/s². 2 kg → 2 m/s².

STEP 1 · COMPARE	STEP 2 · PRODUCT	STEP 3 · SAY IT
mass × ____ · a × ____	1 × 4 = ____ · 2 × 2 = ____	"More mass, ____ acceleration."

4 · EXPLAIN IT · CHOOSE A FRAME

- ☐ My claim is ____ because my data shows ____ .
- ☐ One control was ____ so the test was fair.
- ☐ A trial that did not fit was ____ because ____ .

I said it ☐ · I wrote it ☐

S3 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 11 · S3 · CHOICE BOARD

Mass lab tic-tac-toe

11 D

DIRECTIONS · Complete three in a row. Shade each square you finish.

CALCULATE v rises 0 → 4 m/s in 2 s. Find a.	**DRAW** Sketch two v–t lines: light cart and heavy cart, same force.	**EXPLAIN** Why must the hanging mass stay the same today?
REAL WORLD Why is a loaded truck slower to speed up than an empty one?	**CHALLENGE ★** m × a ≈ 4 N in every row. Predict a for a 5 kg system.	**CALCULATE** 1 kg: 4 m/s². Predict a at 2 kg with the same force.
EXPLAIN What counts as "system mass" in the hanger setup?	**DRAW** Plot a vs. m for 1, 2, 4 kg at 4, 2, 1 m/s².	**REAL WORLD** 2026-XJ has a huge mass. What does that mean for a push?

Three in a row: _____ · The square that taught me most: _____

S3 1 2 3

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Settle the lab argument

11 I

DIRECTIONS · Shade the bubble of the person who is right. Use lab evidence.

ROUND 1

ANA
"We added mass to the cart AND the hanger, so we tested mass fairly."

BEN
"Two things changed. Keep the hanging mass fixed."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"The cart is the only mass that accelerates."

LU
"The hanger accelerates too — count it in the system mass."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"At 1 kg we got 4 m/s², at 2 kg 2 m/s². Mass doesn't matter."

RAJ
"Doubling mass halved a. Mass matters a lot."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

S3 1 2 3

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Comic: the heavy cart is slower to change

11 CS

YOUR READER

A friend who thinks an empty cart and a loaded cart speed up the same with the same push.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **ORANGE** — Equal-length push arrows in panels 1 and 2
- ☐ ◌ **GREEN** — Both carts, because both speed up
- ☐ ◌ **PURPLE** — Panel 3 velocity arrows: longer for the empty cart
- ☐ ◯ **PENCIL** — Each cart's mass in kg

1 · SAME PUSH *An empty cart gets a push.*

CAPTION _____

2 · SAME PUSH *A loaded cart gets the same push.*

CAPTION _____

3 · ONE SECOND LATER *Compare how fast each cart is going.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **mass · force · acceleration · inversely**

_____

_____

S3 1 2 3

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Doodle Notes · fair test

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Fair test · prueba controlada (ES) · teste controlado (PT)



stays the same

TODAY'S CASE

Two carts with masses 1 kg and 2 kg each experience the same 4 N net force.

01 · SKETCH AND LABEL

Draw both carts with equal orange net-force arrows. Label the variable you changed and the quantity you kept fixed.

02 · CHANGE THE PICTURE

Draw how their purple velocities change after one second from rest. Explain which comparison tests the effect of mass.

Annotate your sketches. Circle the mark that best shows fair test. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

I can compare the two accelerations because …

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Frayer Model · fair test

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Fair test · prueba controlada (ES) · teste controlado (PT)

A · A second trial changes both the net force and the mass.

B · Two trials differ in mass while net force stays fixed.

01 · DEFINITION

What does fair test mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

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Picture Dictionary · fair test

I can connect each word to a picture and use it to explain this case.

Name: _____ My home language: _____



mass

masa · massa

amount of matter (kg)

My word or label: _____



control

control · controle

what stays the same

My word or label: _____

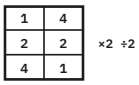


trial

prueba · ensaio

one run of the test

My word or label: _____

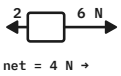


pattern

patrón · padrão

what repeats in data

My word or label: _____



net force

fuerza neta · força resultante

all forces on the cart combined

My word or label: _____



acceleration

aceleración · aceleração

change in velocity per second

My word or label: _____

MATCH · WRITE THE ENGLISH WORD UNDER EACH PICTURE



_____



_____



_____

Use two words to explain the case: Two carts with masses 1 kg and 2 kg each experience the same 4 N net force.

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else